🔥 Flare Network Design — Practical Study Guide

> A field-oriented reference covering the core engineering topics in flare network design — combining API/EPA theory with worked sample calculations, sample process datasheets, and design-basis assumptions drawn from real project execution (process design, hydraulic sizing, mechanical/safety review, and commissioning).

Illustrative project used throughout this guide: a single PSV protecting a propane storage/process vessel, exposed to an external pool-fire scenario, relieving through a horizontal knock-out drum to an elevated steam-assisted flare tip. All numbers below are worked sample calculations for study purposes — always replace with project-specific data.

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1. Design Basis & Assumptions

Every calculation sheet in Section 8 traces back to this basis. In real projects, this section is issued as a standalone **"Flare System Design Basis"** document and frozen before detailed engineering starts — changing it mid-project is one of the biggest sources of rework.

1.1 Site & Ambient Conditions

Parameter	Value	Notes
Ambient temperature (design max)	45 °C (113 °F)	Governs relieving temperature credit, dispersion modeling
Ambient temperature (min)	5 °C (41 °F)	Governs low-temperature material selection, viscosity checks
Design wind speed	5 m/s (base case) & 10 m/s (sensitivity)	Used in flame tilt / radiation footprint runs
Elevation / atmospheric pressure	Sea level, 101.325 kPa	Adjust for inland/high-altitude sites
Solar radiation background	1.0 kW/m² (per API 521)	Added to flare-only radiation total

1.2 Codes & Standards Basis

- API RP 521 (7th Ed.) — relief load, hydraulics, radiation methodology
- API STD 537 — flare tip & assist system design
- API STD 520 Parts I & II — PSV sizing
- ISO 23251 — used where contractually required in lieu of/alongside API 521
- 40 CFR 60.18 — combustion efficiency, minimum heating value, exit velocity
- Energy Institute AIV Guidelines (2nd Ed.) — vibration screening
- Company/client engineering specification (governs where stricter than the above, e.g., house K-factors, minimum pilot count)

1.3 Key Design Assumptions (typical, to be confirmed per project)

Assumption	Typical Value	Basis / Justification
PSV backpressure allowable	10% of set pressure (conventional PSV)	API 520 Part I
Fire case environmental factor, F	1.0 (bare vessel) / 0.3 (with fireproofing)	API 521 Table on environmental factors
Adequate drainage & firefighting credit	Assumed available unless stated otherwise	Reduces fire case factor — confirm with site HSE
KOD droplet removal size	300–600 µm	Flare service; finer cut not justified for flare KODs
KOD liquid residence time	20–30 min (min. to alarm response)	Company philosophy; confirm with Ops
Minimum LHV for flare gas	300 Btu/scf (unassisted) / 200 Btu/scf (assisted)	40 CFR 60.18
Maximum allowable radiation at grade (personnel, escape route)	4.73 kW/m² (1500 Btu/hr·ft²)	API 521
Maximum radiation at equipment (no personnel)	9.46 kW/m²	API 521
Simultaneous relief grouping	By common initiating cause only	API 521 Section 4
Corrosion allowance (KOD, CS)	3 mm	Project material spec
Design life	25 years	Project spec

> ⚠️ **Practical note:** Always issue the design basis for client/PMC approval *before* running detailed hydraulics — changing K-factors or the fire-case environmental factor after headers are sized is one of the most common (and expensive) causes of rework on brownfield flare projects.

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2. Gas Characterization

2.1 Flow Rate & Variability

- **Normal relief** — routine venting/blowdown; sizes minimum sustained tip turndown.

- **Maximum relief (design case)** — fire, power failure, blocked outlet, control valve failure; governs header/sub-header diameter.
- **Staged/rundown relief** — sequential PSV lifting during upset; needs dynamic simulation for complex multi-source headers.

Practical experience: Build a PSV relief load summary (Section 9.1) as the single source of truth before touching hydraulics. Identify the governing scenario **per header segment**, not globally — a common error is applying one global governing case across the whole network.

2.2 Composition & Heating Value

- MW, specific gravity, and LHV drive flare tip selection, combustion efficiency compliance, and smoke potential.
- Use a *flow-weighted average composition* across scenarios, not just the largest-flow case — a small hydrogen-rich stream can dominate combustion behavior.

2.3 Two-Phase Relief Scenarios

- Use the **Omega method (API 520 Appendix)** or **HEM** for two-phase PSV sizing — don't default to vapor-only sizing for flashing streams.
- Two-phase lines carry much higher density than vapor-only lines — this changes header sizing significantly and often justifies a dedicated sub-header to the KOD.

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3. Flare System Hydraulics

3.1 Header & Sub-Header Sizing

Sizing sequence used in practice:

1. Build routing sketch (tie-ins, lengths, elevations).
2. Run hydraulic calc (friction + elevation loss, Mach check) from each PSV to the tip.
3. Check simultaneous relief cases per the cause-and-effect matrix — not just the single largest contributor.
4. Iterate pipe size until **every** PSV tie-in backpressure is within its allowable limit — the tightest constraint is often a low-set-pressure PSV far from the tip, not the largest PSV.

3.2 Mach Number & Velocity Limits

Parameter	Typical Limit	Why
Mach number, header (design)	≤ 0.7, ≤ 0.5 preferred	Avoid choking, excess ΔP
Mach number, short PSV tailpipe	up to 1.0 (sonic) acceptable	Short-duration transient only
Velocity (AIV screening)	Per Energy Institute likelihood ranking	Prevent acoustic-induced vibration fatigue

3.3 Simultaneous Relief Events

- Group by **common initiating cause** (power failure, cooling water failure, fire zone) — only sources triggered by the same event are summed.
- Fire zone cases: only PSVs on equipment within the same fire zone (commonly a 232 m^2 / 2,500 ft^2 radius per API 521) are counted together.
- Build a cause-and-effect matrix first — treating every PSV's individual worst case as simultaneous massively over-sizes the header.

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4. Knock-Out Drum (KOD) Design

4.1 Liquid Separation — Souders-Brown Equation

```
V_max = K_SB × √[(ρ_L − ρ_V) / ρ_V]
```

`K_SB` depends on target droplet size (300–600 µm typical for flare KODs — coarser than process separators, since over-designing to a fine cut like 150 µm gives an oversized drum with little real benefit in flare service).

4.2 Residence Time & Settling Velocity

- Liquid space sized on residence time (20–30 min typical, per site philosophy) for operator response before carryover.
- Always cross-check against the **maximum liquid inventory/slug scenario** — residence time alone can under-size the drum if a slug governs rather than continuous flow.

4.3 Horizontal vs. Vertical KODs

Type	Pros	Cons	Typical Use
Horizontal	Better for high liquid loading, longer settling length, easier debottlenecking	Larger footprint	Refineries, high carryover risk
Vertical	Smaller footprint	Limited holdup, less efficient at high flow	Offshore/FPSO, space-constrained sites

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5. Flare Tip & Pilot Design

5.1 Types of Flare Tips

Type	Mechanism	Best For	Trade-off
Utility (basic)	No assist	Low-value, intermittent, remote flares	Poor smoke control at high loads
Steam-assist	Steam-induced air entrainment	Refineries/petrochemical (steam available)	Over-steaming quenches flame/steam plume
Air-assist	Forced-draft blower	Sites without reliable steam	Blower power & maintenance
Gas-assist	Sonic gas jets induce air entrainment	Remote/unmanned sites	Continuous fuel gas consumption

Practical tip: Steam ratio should be tuned during commissioning against real (not just design-case) relief loads — over-injection is one of the most common field operability complaints.

5.2 Pilot Systems

- Continuously lit, monitored per **EPA 40 CFR 60.18**; requires reliable ignition (flame-front generator or high-energy igniter) and flame detection (thermocouple or UV/IR) alarmed to the control room.
- Minimum 2 (commonly 3) redundant pilots is standard; single-pilot designs are a frequent HAZOP/PHA finding. Pilot gas supply should be independently regulated from the main flare gas supply to avoid common-mode failure.

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6. Radiation & Safety Analysis

6.1 Thermal Radiation Limits (API RP 521)

Radiation Level (kW/m²)	Exposure Condition
1.6	No shielding, indefinite exposure
4.73	Emergency actions, several minutes exposure, escape route available
6.31	Emergency actions lasting only seconds
9.46	Equipment area, personnel excluded during flaring

Practical experience: Run radiation contours for multiple flow cases (normal, intermediate, max) and multiple wind directions/speeds — flame lift/tilt due to wind materially shifts the footprint. Always add the 1.0 kW/m² solar background allowance — a commonly missed detail.

6.2 Stack Height & Dispersion Modeling

- Governed by the larger of: radiation limits at grade/nearest equipment, or ground-level concentration limits for **unignited** release (e.g., pilot failure during a relief event).
- Run dispersion modeling (e.g., AERMOD) for the unignited case in addition to the ignited radiation case, especially for H_2S-bearing streams — this can govern stack height.

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7. Environmental & Regulatory Compliance

7.1 API Standards

- **API 521** — relief load determination, header sizing philosophy, radiation/dispersion methodology.
- **API 537** — flare tip mechanical/combustion design, assist systems, pilot requirements.

7.2 ISO 23251

Internationally harmonized equivalent of API 521, used where contractually specified (common on European/Middle East/multinational projects).

7.3 EPA Requirements (40 CFR 60.18)

- ⩾98% combustion efficiency, demonstrated via: LHV ⩾ 300 Btu/scf (or ⩾200 Btu/scf assisted), exit velocity limits per the regulation's Btu/scf-vs-velocity chart, and no visible emissions >5 min in any 2 consecutive hours.

- **Practical note:** Continuous flare gas flow/composition monitoring (GC or NIR analyzer on the header) is increasingly required under site-specific consent decrees — always check current permit conditions, which often exceed the 40 CFR 60.18 baseline.

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8. Sample Calculation Sheets

> All calculations below use the illustrative project basis in Section 1. Values are for study purposes — verify against project-specific PSV datasheets and simulation results.

8.1 Calc Sheet 1 — Fire Case Relief Load (API 521 Wetted Area Method)

Given:

- Vessel wetted surface area (fire exposure), `A` = 1,000 ft²
- Environmental factor, `F` = 1.0 (bare vessel, adequate drainage assumed)
- Latent heat of vaporization at relieving conditions, `L` = 150 Btu/lb (propane, approx.)

Step 1 — Total absorbed heat (API 521 Eq., A ≤ 2,800 ft²):

```
Q = 21,000 × F × A^0.82
Q = 21,000 × 1.0 × (1,000)^0.82
Q = 21,000 × 288.4
Q ≈ 6,056,000 Btu/hr  (≈ 6.06 MMBtu/hr)
```

Step 2 — Relieving mass flow:

```
W = Q / L
W = 6,056,000 / 150
W ≈ 40,373 lb/hr   (≈ 18.3 t/hr, ≈ 18,300 kg/hr)
```

Result: Governing fire-case relief load ≈ **40,373 lb/hr** of propane vapor. This becomes the design flow for the PSV, header segment, and downstream KOD/tip sizing below.

> 📌 **Assumption check:** If fireproofing is credited, F drops to ~0.3 and Q/W drop proportionally — confirm fireproofing coverage and rating with the mechanical/piping team before applying a reduced F.

8.2 Calc Sheet 2 — Header Hydraulics & Mach Number Check

Given (carried from Calc Sheet 1):

- Mass flow, `W` = 40,373 lb/hr
- Relieving temperature, `T` = 760 °R (300 °F)
- Header pressure at this point, `P` = 29.7 psia (15 psig)

- MW = 44 (propane), k = 1.13
- Trial pipe size: 10-inch, Schedule 40 (ID = 10.02 in)

Step 1 — Gas density at relieving conditions (ideal gas):

```
ρ = (P × MW) / (R × T),    R = 10.73 psia·ft³/(lbmol·°R)
ρ = (29.7 × 44) / (10.73 × 760)
ρ = 1,306.8 / 8,154.8
ρ ≈ 0.160 lb/ft³
```

Step 2 — Volumetric flow:

```
Q_v = W / ρ = 40,373 / 0.160 ≈ 252,300 ft³/hr ≈ 70.1 ft³/s
```

Step 3 — Pipe flow area (10-in Sch 40):

```
ID = 10.02 in = 0.835 ft
A = (π/4) × (0.835)² = 0.548 ft²
```

Step 4 — Actual velocity:

```
V = Q_v / A = 70.1 / 0.548 ≈ 127.9 ft/s
```

Step 5 — Speed of sound:

```
c = √[k × (1,545/MW) × T × 32.2]
c = √[1.13 × (1,545/44) × 760 × 32.2]
c = √[1.13 × 35.11 × 760 × 32.2]
c ≈ 985.5 ft/s
```

Step 6 — Mach number:

```
M = V / c = 127.9 / 985.5 ≈ 0.13
```

Result: M = 0.13, well under the 0.5–0.7 design limit. **10-inch header is acceptable** for this segment on velocity/Mach grounds — backpressure at each upstream PSV tie-in must still be separately verified (Calc Sheet not shown here; use standard Darcy-Weisbach/isothermal compressible flow with header fittings K-values).

> 📌 **Practical note:** This low Mach number leaves margin — on a real project, check whether a smaller (e.g., 8-inch) header still meets Mach/backpressure limits before

> locking in 10-inch, since header size is a major cost driver.

8.3 Calc Sheet 3 — Knock-Out Drum Sizing (Horizontal, Souders-Brown)

Given (carried from Calc Sheets 1–2):

- Vapor mass flow, `W_v` = 40,373 lb/hr
- Vapor density, `ρ_V` = 0.160 lb/ft³
- Liquid density, `ρ_L` = 30 lb/ft³ (light HC liquid at relieving conditions)
- Droplet cut / `K_SB` = 0.20 ft/s (flare service, coarse cut, no mist eliminator — plugging risk)
- Assumed liquid carryover rate for holdup case = 5,000 lb/hr
- Required liquid residence time = 20 min

Step 1 — Maximum allowable vapor velocity (Souders-Brown):

```
V_max = K_SB × √[(ρ_L − ρ_V) / ρ_V]
V_max = 0.20 × √[(30 − 0.16) / 0.16]
V_max = 0.20 × √(186.5)
V_max = 0.20 × 13.66
V_max ≈ 2.73 ft/s
```

Step 2 — Vapor volumetric flow:

```
Q_v = W_v / ρ_V = 40,373 / 0.160 = 252,300 ft³/hr = 70.1 ft³/s
```

Step 3 — Required vapor flow area (horizontal drum, ~50% cross-section available above liquid level):

```
A_vapor = Q_v / V_max = 70.1 / 2.73 ≈ 25.7 ft²
A_vapor ≈ 0.5 × (π/4) × D²
25.7 = 0.5 × 0.785 × D²
D² = 25.7 / 0.3925 = 65.5
D ≈ 8.09 ft  →  select D = 8 ft-0 in ID
```

Step 4 — Drum length (L/D = 4, typical for flare KOD):

```
L = 4 × D = 4 × 8 = 32 ft (tan-to-tan, to be confirmed with heads/nozzle layout)
```

Step 5 — Liquid holdup volume check (20-min residence, slug scenario):

```
Q_liquid = 5,000 lb/hr / 30 lb/ft³ = 166.7 ft³/hr = 2.78 ft³/min
V_holdup = 2.78 × 20 = 55.6 ft³
```

Cross-check: available shallow-layer volume in an 8 ft × 32 ft drum (e.g., 6-in liquid layer) ≈ 8 × 32 × 0.5 = 128 ft^3 ≫ 55.6 ft^3 required → **liquid holdup is not governing**; vapor/droplet removal governs the drum diameter.

Result: Horizontal KOD, **8'-0" ID × 32'-0" T/T**, confirmed adequate for both droplet removal and liquid holdup for this case.

> 📌 **Assumption check:** K_SB of 0.20 ft/s is a common flare-service value but varies by company philosophy (0.15–0.35 ft/s range seen in practice) — confirm against project specification before finalizing drum diameter, since D scales with √(1/K_SB).

8.4 Calc Sheet 4 — Radiation Distance Check (API 521 Point-Source Method)

Given:

- Total heat release at flare tip, `Q` = 100 MMBtu/hr (illustrative combined/governing case, larger than the single fire case above)
- Fraction of heat radiated, `τ` (tau) = 0.30 (typical for hydrocarbon flares, range 0.2–0.4 depending on assist type/smokeless performance)
- Target radiation level at grade, `K` = 1,500 Btu/hr·ft^2 (≈ 4.73 kW/m^2, personnel with escape route)

Step 1 — Rearranged point-source equation:

```
K = (τ × Q) / (4π D²)
→ D = √[τ × Q / (4π × K)]
```

Step 2 — Substitute values:

```
D = √[(0.30 × 100,000,000) / (4 × 3.1416 × 1,500)]
D = √[30,000,000 / 18,850]
D = √1,591.5
D ≈ 39.9 ft ≈ 12.2 m
```

Result: Minimum slant distance from flame center to any point where personnel may need to take emergency action (with escape route) ≈ **12.2 m**. Repeat with K = 9.46 kW/m^2 (3,000 Btu/hr·ft^2) for equipment-only zones, and re-run with flame tilt geometry (wind-adjusted flame center offset) for the actual stack-height/plot-plan check — the simple point-source method above ignores flame tilt and is typically only a first-pass screening calc; detailed radiation studies use multi-point source or API 521 Annex flame-tilt methodology.

> 📌 **Assumption check:** τ (fraction radiated) is highly sensitive to flare type — steam-assisted smokeless flares are often modeled nearer 0.2, while unassisted/sooty flames can approach 0.4. Confirm the project's basis value; it materially changes the required safe distance / stack height.

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9. Sample Datasheets

9.1 PSV / Relief Load Summary Datasheet

Tag No.	Service	Set Pressure (barg)	Relieving Scenario	Relieving Temp (°C)	Mass Flow (kg/hr)	MW	Phase	k	Governing?
PSV-101	Propane Storage Vessel	17.5	External Fire	149	18,300	44.1	Vapor	1.13	✅ Yes
PSV-102	Propane Storage Vessel	17.5	Blocked Outlet	65	9,100	44.1	Vapor	1.13	No
PSV-201	Deethanizer OVHD	24.0	Power Failure	55	12,400	32.6	Vapor	1.20	✅ Yes (segment 2)
PSV-301	Reflux Drum	8.5	Control Valve Failure	40	6,700	38.2	Two-Phase	1.18	No

(Illustrative — a real relief load summary carries every PSV in the unit, cross-referenced to the cause-and-effect matrix and P&ID tag.)

9.2 Knock-Out Drum (KOD) Process Datasheet

Parameter	Value	Unit
Tag No.	V-501	—
Service	Flare Knock-Out Drum	—
Orientation	Horizontal	—
Inside Diameter	2,438 (8'-0")	mm
Tangent-to-Tangent Length	9,754 (32'-0")	mm
Design Pressure	3.5 (full vacuum to +3.5)	barg
Design Temperature	−29 to 150	°C
Operating Pressure (normal)	0.15	barg
Governing Relief Case	Fire Case (Vessel PSV-101)	—
Vapor Inlet Flow (governing case)	18,300	kg/hr
Vapor Density (relieving cond.)	2.56	kg/m^3
Liquid Density (relieving cond.)	480	kg/m^3
Droplet Removal Size (design)	400	µm
K_SB (Souders-Brown constant)	0.061	m/s (≈0.20 ft/s)
Max. Allowable Vapor Velocity	0.83	m/s
Liquid Residence Time (design)	20	min

Parameter	Value	Unit
High Level Alarm	50%	% of ID
High-High Level Trip (ESD)	65%	% of ID
Corrosion Allowance	3	mm
Material of Construction (shell)	SA-516 Gr.70	—
Internals	Inlet deflector/vane, no mist eliminator (plugging risk)	—
Nozzles	Inlet (1), Vapor outlet to flare (1), Liquid outlet/pump-out (1), Drain, Level instruments (LG/LT), PSV (thermal relief), Manway	—
Applicable Code	ASME Section VIII Div. 1	—

9.3 Flare Tip Datasheet

Parameter	Value	Unit
Tag No.	FT-701	—
Type	Steam-Assisted, Smokeless	—
Tip Diameter	610 (24-in)	mm
Riser Height (grade to tip)	45	m
Design (Max.) Relief Flow	45,000	kg/hr
Normal/Continuous Purge Flow	150	kg/hr
Minimum LHV for Smokeless Operation	300 (unassisted basis)	Btu/scf
Exit Velocity at Max. Flow	0.5 Mach (≈170)	m/s
Steam-to-Gas Ratio (design)	0.3 : 1 (mass basis)	—
Steam Supply Pressure	10	barg
Turndown Ratio	20:1	—
Pilot Burners	3 (2 required + 1 standby)	—
Pilot Ignition System	Flame Front Generator (FFG)	—
Pilot Monitoring	Thermocouple + UV/IR scanner, alarmed to DCS	—
Molecular Seal / Purge Gas	N_2, velocity seal, 0.05 m/s min. purge	—
Design Combustion Efficiency	⩾ 98%	—
Applicable Standard	API STD 537, 40 CFR 60.18	—
Radiation at Grade (design case, 12.2 m)	4.73	kW/m^2

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10. Practical Design Checklist

- [ ] Design basis issued and approved (Section 1) before hydraulics begin
- [ ] PSV relief load summary complete (all scenarios, phases, compositions)

- [ ] Governing case identified **per header segment**, not just globally
- [ ] Two-phase relief cases flagged and sized separately (Omega/HEM method)
- [ ] Simultaneous relief cause-and-effect matrix built
- [ ] Backpressure verified at every PSV tie-in (including existing PSVs on brownfield tie-ins)
- [ ] Mach number and AIV screening performed on all high-velocity segments
- [ ] KOD droplet size / K_SB basis agreed with client/site philosophy
- [ ] KOD liquid residence time checked against worst-case slug/dump scenario (see Calc Sheet 3)
- [ ] Flare tip assist type selected based on utility availability (steam/air/fuel gas)
- [ ] Pilot redundancy and independent gas supply confirmed
- [ ] Radiation contours run for multiple flow cases and wind conditions (see Calc Sheet 4)
- [ ] Solar radiation background included in radiation totals
- [ ] Unignited dispersion case modeled for stack height determination
- [ ] Combustion efficiency basis (LHV, exit velocity) checked against 40 CFR 60.18 / local regulation
- [ ] Site-specific consent decree/permit conditions cross-checked
- [ ] All datasheets (PSV summary, KOD, flare tip) issued for vendor/EPC use

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11. Common Field Issues & Lessons Learned

Issue	Root Cause	Practical Fix
Excessive visible smoke at partial loads	Steam ratio not tuned for real (vs. design) flow	Re-tune steam-to-gas ratio during commissioning across turndown range
AIV fatigue cracking at small-bore branches	Under-screened high-velocity segments	Retrofit reinforced branches or reroute instrument connections away from high-Mach zones
Liquid carryover to flare tip	KOD undersized for slug/dump scenario, or ESD trip untested	Revisit liquid residence time basis + verify trip testing frequency
Nuisance pilot-out trips	Poor flame detector line-of-sight or wind-affected pilot design	Reposition scanners, consider wind-shielded pilot tip
Backpressure exceedance after brownfield tie-in	New PSV added without re-verifying network-wide hydraulics	Re-run full hydraulic model network-wide for any new tie-in
Radiation exceedance during unexpected high flaring	Radiation study only covered single "worst case"	Model radiation across a range of flows; publish safe-distance curves for Ops
KOD oversized/costly	Over-conservative K_SB (fine droplet cut applied unnecessarily)	Confirm K_SB basis against company philosophy before sizing (see Calc Sheet 3 note)

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12. Case Study — Brownfield LPG Sphere Relief Tie-In

> A composite, illustrative case study based on the type of finding commonly encountered during brownfield flare network debottlenecking at refineries/terminals. Names, tag numbers, and figures are representative, not project-specific.

12.1 Background

A terminal was adding a new LPG storage sphere (V-410, MAWP 250 psig) with its own PSV (PSV-410) tied into an existing plant flare header that already served four other PSVs, including an older low-set-pressure PSV (PSV-115, set at 15 psig) on a distillation column overhead drum located far downstream on the same sub-header. The tie-in was engineered as a "local" addition: the new PSV-410 line was sized and hydraulically checked against the *nearest* header segment only, on the assumption that the existing network had adequate spare capacity.

12.2 Problem Identified

During detailed engineering hydraulic re-verification (triggered by the project's MOC/PSSR process, not the original tie-in design), the engineer re-ran the **network-wide** backpressure model with PSV-410 relieving simultaneously with the governing fire-zone group that included PSV-115. Two issues surfaced:

1. **Backpressure exceedance at PSV-115:** With the new PSV-410 flow added to the shared header, calculated backpressure at PSV-115's outlet rose to **14% of its set pressure**, exceeding the 10% allowable limit for its conventional (non-balanced) trim — even though PSV-410's own local tie-in point was well within limits. The governing constraint was **not** at the new equipment at all, but at an existing, unrelated PSV far downstream — exactly the brownfield pitfall flagged in this guide's Section 3.1 practical note.
2. **KOD liquid loading:** The existing KOD (V-405) had originally been sized years earlier using a fine droplet-cut basis ($K\_SB \approx 0.10$ ft/s) inherited from an old company standard. When the current project's engineer re-ran the Souders-Brown check (per the Calc Sheet 8.3 method in this guide) using the current corporate K_SB of 0.20 ft/s, the drum was confirmed to have **more real margin** than the original design assumed — but this had never been checked, so the project team had initially budgeted for a costly KOD replacement that turned out to be unnecessary.

12.3 Investigation & Recalculation

- **Network hydraulics:** The team rebuilt the full hydraulic model (not just the local tie-in segment) using the Calc Sheet 8.2 methodology of this guide, applied at every PSV tie-in point rather than only the new one. This confirmed PSV-115 as the governing constraint and identified two viable fixes: (a) upsize ~120 m of the shared sub-header from 8-inch to 10-inch, or (b) reroute PSV-410 into a separate sub-header joining the main header downstream of PSV-115's tie-in point.
- **KOD re-verification:** Using the current K_SB = 0.20 ft/s basis (Calc Sheet 8.3 method) against V-405's actual as-built diameter and the updated combined relief load (including PSV-410), the drum was confirmed adequate without modification — avoiding an unnecessary replacement.
- **Radiation re-check:** Because the combined relief load changed, radiation at the two nearest occupied work areas was re-run (Calc Sheet 8.4 method) for the new simultaneous case. The result remained within the 4.73 kW/m^2 limit with margin, so no stack height change was required.

12.4 Root Cause

Two compounding root causes were identified:

1. **Local-only hydraulic verification at tie-in design —** the new PSV-410 line was checked against its immediate connection point but the shared network was never re-run end-to-end for the new simultaneous relief case, so the impact on PSV-115 (a completely different, older PSV) was missed until the MOC review.
2. **Undocumented legacy design margin at the KOD —** the original K_SB basis used for V-405 was never recorded in a retrievable design basis document, so the project defaulted to assuming the drum needed replacement rather than re-verifying it against the current, less conservative, corporate standard.

12.5 Resolution

- **Hydraulics:** Option (b) was selected — PSV-410 was rerouted into a new, independent sub-header tying into the main header downstream of PSV-115, avoiding both the sub-header upsize cost and any backpressure impact on the existing PSV. This is generally the lower-cost fix on brownfield headers when a downstream tie-in point is geometrically available.
- **KOD:** No hardware change required; the re-verified Souders-Brown calculation was issued as an updated design basis record for V-405 so future tie-in projects would not repeat the "assume replacement needed" default.
- **Radiation:** No change required; updated contour issued for records.
- **Process safety management:** The terminal's MOC procedure was updated to require a network-wide hydraulic re-run (all existing PSV tie-ins, not just the new one) as a mandatory checklist item for any new flare header tie-in, however small the new load appears relative to total header capacity.

12.6 Outcome

- The rerouted sub-header addition was implemented during the original construction window with a minor (2-week) engineering schedule impact, avoiding a much larger cost and schedule hit that a full 120 m sub-header upsize (or unnecessary KOD replacement) would have caused.
- The finding was documented as a corporate lessons-learned item: "small" brownfield tie-ins require the same network-wide verification rigor as new grassroots headers, since the governing constraint is frequently at an unrelated, pre-existing PSV rather than at the new connection itself.

12.7 Case-Specific Lessons Learned

Lesson	Applied Fix
A "local" tie-in check is not sufficient — the governing backpressure constraint is often at an unrelated, pre-existing PSV elsewhere on the shared header	Mandate network-wide (all PSV tie-ins) hydraulic re-verification for every new flare header connection, regardless of how small the added load seems
Undocumented legacy design margins (e.g., K_SB basis) lead to costly default assumptions	Record and retain the design basis (K_SB, residence time, etc.) for every KOD so future projects can re-verify rather than assume replacement is needed
Rerouting into a downstream tie-in point can be a lower-cost fix than upsizing a shared sub-header	Evaluate routing alternatives before defaulting to pipe upsizing when a backpressure exceedance is found
Radiation and KOD checks should be re-run whenever the combined relief load changes, even if the new PSV's own load is small	Include radiation and KOD re-verification explicitly in the brownfield tie-in MOC checklist, not just hydraulics

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13. Reference Standards

- **API RP 521** — Pressure-relieving and Depressuring Systems
- **API STD 537** — Flare Details for Petroleum, Petrochemical, and Natural Gas Industries
- **API STD 520** (Parts I & II) — Sizing, Selection, and Installation of Pressure-relieving Devices
- **ISO 23251** — Petroleum, petrochemical and natural gas industries — Pressure-relieving and depressuring systems
- **40 CFR 60.18** — General Control Device and Work Practice Requirements (US EPA)
- **Energy Institute Guidelines for the Avoidance of Vibration Induced Fatigue Failure** — AIV screening methodology

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This guide is a practical study reference combining standard design theory with worked sample calculations and lessons learned from flare network design/review experience. All numeric examples are illustrative — always validate against project-specific design basis, vendor data, simulation results, and current regulatory requirements.